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Wheel assembly system and method

Abstract

A system comprises a docking station and a robot. The docking station includes a first dock and a second dock. The first dock is configured to removably-receive a first plate. The first plate is configured to removably-receive a first plurality of parts. The second dock is configured to removably-receive a second plate. The second plate is configured to removably-receive a second plurality of parts. The robot includes an end effector configured to engage the first plate and the second plate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/421,316 filed Nov. 1, 2022, the disclosure of which is considered part of the disclosure of this application and is incorporated herein by reference in its entirety.

TECHNICAL FIELD
(1) This disclosure relates generally to a system and method for assembling a wheel, and more particularly to a dual-staged system and method for assembling a wheel.

BACKGROUND

(2) This section provides background information related to the present disclosure and is not necessarily prior art. Wheel assemblies may include a wheel, a valve stem and/or a tire pressure monitoring system (TPMS) sensor. Known wheel assembly systems include (i) a tool that engages a valve stem and assembles the valve stem through an aperture formed in a wheel, and (ii) another tool that engages a TPMS sensor and assembles the TPMS sensor through an aperture formed in a wheel. Accordingly, known wheel assembly systems may include a first station for staging a valve stem for receipt by a first tool, and a second station for staging receipt of a TPMS sensor for receipt by a second tool. In known methods of assembling a wheel, the first and second tools may operate to receive the valve stem and the TPMS sensor, respectively, and install the valve stem and the TPMS sensor relative to the wheel. While known wheel assembly systems and methods have proven acceptable for their intended purposes, a continuous need for improvement in the pertinent art exists.

SUMMARY

(3) This section provides a general summary of the disclosure, and is not a comprehensive disclosure of its full scope or all of its features.

(4) One aspect of the disclosure provides a system. The system comprises a docking station and a robot. The docking station includes a first dock and a second dock. The first dock is configured to removably-receive a first plate. The first plate is configured to removably-receive a first plurality of parts. The second dock is configured to removably-receive a second plate. The second plate is configured to removably-receive a second plurality of parts. The robot includes an end effector configured to engage the first plate and the second plate.

(5) Implementations of this aspect of the disclosure may include one or more of the following optional features. In some implementations, the first dock includes a first attachment fixture and a second attachment fixture. The first attachment fixture may be configured to receive a first part of the first plurality of parts. In some implementations, the second attachment fixture is configured to receive a second part of the first plurality of parts.

(6) In some implementations, the first dock includes a base and a pair of arms extending from the base. The first dock may include a sensor coupled to the base and configured to sense a location of the first part.

(7) In some implementations, the end effector includes a first engagement member and a second engagement member. The first engagement member may be configured to engage the first plate. The second engagement member may be configured to engage the second plate when the first engagement member engages the first plate.

(8) In some implementations, the system comprising the first plate includes a first peripheral tab and a second peripheral tab. The first dock may include a first attachment fixture and a second attachment fixture. The first attachment fixture may be configured to receive the first peripheral tab. The second attachment fixture may be configured to receive the second peripheral tab. In some implementations, the robot further comprises a first alignment device and a second alignment device. The first alignment device may oppose the first peripheral tab. The second alignment device may oppose the second peripheral tab.

(9) In some implementations, the first plurality of parts includes a first type of part, and the second plurality of parts includes a second type of part different than the first type of part.

(10) Another aspect of the disclosure provides a method. The method may include positioning, with a first robot, a first part on a first plate disposed at a docking station. The method may also include securing a second robot to a second plate disposed at the docking station. The method may further include installing, with the second robot, a second part on a wheel. The method may also include releasing the second robot from the second plate at the docking station. The method may further include securing the second robot to the first plate at the docking station.

(11) Implementations of this aspect of the disclosure may include one or more of the following

optional features. In some implementations, installing, with the second robot, the second part on the wheel is simultaneous with positioning, with the first robot, the first part on the first plate.
(12) In some implementations, securing the first plate to the second robot is simultaneous with releasing the first plate from the second robot.
(13) In some implementations, installing, with the second robot, the second part on the wheel includes applying, with the second robot, a first force in a first direction on the wheel, and applying, with the second robot, a second force in a second direction on the wheel, wherein the second direction is opposite the first direction.
(14) In some implementations, the method further comprises sensing a location of the first part.
(15) In some implementations, the method further comprises installing, with the second robot, a third part on the wheel. The method may further comprise simultaneously (i) positioning, with the first robot, a fourth part on the first plate disposed at the docking station and (ii) installing, with the second robot, the third part on the wheel.
(16) In some implementations, the first part is a first type of part, and the second part is a second type of part different than the first type of part.
(17) Yet another aspect of the disclosure provides a docking assembly. The docking assembly includes a first dock, a first plate, a second dock, and a second plate. The first plate is removably-disposed on the first dock and includes a first nest and a second nest. The first nest and the second nest are each configured to removably-receive a first part and a second part. The second plate is removably-disposed on the second dock and includes a third nest and a fourth nest. The third nest and the fourth nest are each configured to removably-receive the first part and the second part.
(18) Implementations of this aspect of the disclosure may include one or more of the following optional features. In some implementations, the first plate includes a first peripheral tab and a second peripheral tab, the first nest is disposed on the first peripheral tab, and the second nest is disposed on the second peripheral tab.
(19) In some implementations, the docking assembly further includes a first sensor and a second sensor. The first sensor may be configured to sense a position of the first nest. The second sensor may be configured to sense a position of the second nest.
(20) In some implementations, the first dock includes a first attachment fixture having a first base and a first pair of arms extending from the first base. The first sensor may be disposed on the first base. The second dock may include a second attachment fixture having a second base and a second pair of arms extending from the second base. The second sensor may be disposed on the second base.
(21) The details of one or more implementations of the disclosure are set forth in the accompanying drawings and the description below. Other aspects, features, and advantages will be apparent from the description and drawings, and from the claims.

Description

DESCRIPTION OF DRAWINGS
(1) The drawings described herein are for illustrative purposes only of selected configurations and not all possible implementations, and are not intended to limit the scope of the present disclosure.
(2) FIG. **1** is a perspective view of a wheel assembly system according to the principles of the present disclosure.
(3) FIG. **2** is perspective view of a loading station and a first robot of the wheel assembly system of FIG. **1**, the first robot illustrated in a first orientation according to the principles of the present disclosure.
(4) FIG. **3** is a perspective view of a portion of the loading station and first robot of FIG. **2**.
(5) FIG. **4** is a perspective view of a staging station of the wheel assembly system of FIG. **1** and the

first robot of FIG. **2**, the first robot illustrated in a second orientation according to the principles of the present disclosure.

(6) FIG. **5** is a perspective view of the staging station of FIG. **4** and a second robot of the wheel assembly system of FIG. **1**, the second robot illustrated in a first orientation.

(7) FIG. **6** is a perspective view of an assembly station of the wheel assembly system of FIG. **1** and the second robot of FIG. **5**, the second robot illustrated in a second orientation.

(8) FIG. **7** is a perspective view of the assembly station of FIG. **6** and the second robot of FIG. **5**, the second robot illustrated in a third orientation.

(9) FIG. **8** is a perspective view of the assembly station of FIG. **6** and the second robot of FIG. **5**, the second robot illustrated in the first orientation.

(10) Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

(11) Example configurations will now be described more fully with reference to the accompanying drawings. Example configurations are provided so that this disclosure will be thorough, and will fully convey the scope of the disclosure to those of ordinary skill in the art. Specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of configurations of the present disclosure. It will be apparent to those of ordinary skill in the art that specific details need not be employed, that example configurations may be embodied in many different forms, and that the specific details and the example configurations should not be construed to limit the scope of the disclosure.

(12) Referring to FIG. **1**, a wheel assembly system **10** may be utilized to assemble various components relative to a wheel **12**. In this regard, while the system **10** is generally shown and described herein as being utilized to assembly valve stems **14** and tire pressure sensors **16** relative to the wheel **12**, it will be appreciated that the system **10** may be utilized to assemble, or otherwise provide, other components (e.g., weights, soap, etc.) to the wheel **12**.

(13) The system **10** may include a loading station **18**, a staging station **20**, an assembly station **22**, a first robot **24**, and a second robot **26**. The loading station **18** may include a first hopper system **30**, a second hopper system **32**, and a vision system **34**. The first hopper system **30** may include a first hopper **36**, a first tray **38** disposed downstream of the first hopper **36**, a second tray **40** disposed downstream of the first tray **38**, a container **42** disposed downstream of the second tray **40**, and one or more vibration mechanisms **44**. The second hopper system **32** may include a first hopper **46**, a first tray **48** disposed downstream of the first hopper **46**, the container **42** disposed downstream of the first tray **48**, and the vibration mechanism(s) **44**. As will be explained in more detail below, the first hopper system **30** may receive and convey a first component (e.g., tire pressure sensors **16**) from the first hopper **36** to the container **42** through gravity and/or vibration produced by the vibration mechanism(s) **44**, and the second hopper system **32** may receive and convey a second component (e.g., valve stems **14**) from the first hopper **46** to the container **42** through gravity and/or vibration produced by the vibration mechanism(s) **44**. As illustrated in FIG. **3**, the container **42** may include a divider **50** dividing the container into a first portion **52** that receives the first component and a second portion **54** that receives the second component.

(14) With reference to FIG. **4**, the staging station **20** may include one or more docks **58**. Each dock **58** may include a base **62**, a plate **64**, a robot attachment fixture **66**, one or more part attachment fixtures **67**, one or more nests **68**, and one or more sensors **69**. The plate **64** may be removably-disposed on the base **62** and include a one or more tabs **70** extending from a periphery P of the plate **64**. The attachment fixture **66** may be disposed on the plate **64** and include one or more apertures **72**. Each nest **68** may be disposed on and/or coupled to, the plate **64** (e.g., on a tab **70**) and include one or more recesses **74** configured to receive the part (e.g., tire pressure sensors **16** and/or valve stems **14**).

(15) In some implementations, each dock **58** includes three attachment fixtures **67** coupled to the base **62**. The attachment fixtures **67** may each include a base **78** and a pair of arms **80** extending

from the base **78**. In some implementations, the arms **80** extend orthogonally from the base **78** and define a gap **82** therebetween. In this regard, the arms **80** and the base **78** may collectively define a U-shape. The sensor **69** may include a proximity sensor or any other suitable sensing device operable to sense the presence of a part (e.g., tire pressure sensors **16** and/or valve stems **14**) and/or a nest **68** within the U-shape of a respective attachment fixture **67**. In some implementations, the sensor **69** is coupled to the attachment fixture **67**. For example, the sensor **69** may be coupled to the base **78**.

(16) With reference to FIGS. **7** and **8**, the assembly station **22** may include a frame **84**, a conveyor **86**, one or more alignment pins **88**, and a vision system **90**. The conveyor **86** may be supported by the frame and include one or more rollers **92** to receive the wheel **12**. The rollers **92** may define one or more gaps **94** therebetween. In some implementations, the rollers **92** include a sleeve **95** disposed around the roller. The alignment pins **88** may be disposed within the gaps **94** and collectively define a circular shape. The vision system **90** may be supported by the frame **84** and include a camera **96**.

(17) With reference to FIGS. **3** and **4**, the first robot **24** may include a first arm **101**, a second arm **102**, a third arm **104**, a fourth arm **106**, and an end effector **108**. The first, second, third, and/or fourth arms **101**, **102**, **104**, **106** and/or the end effector **108** may be coupled to, and move (e.g., rotate, pivot, translate, etc.) relative to, one another.

(18) With reference to FIGS. **1** and **5-7**, the second robot **26** may include a first arm **110**, a second arm **112**, a third arm **114**, a fourth arm **116**, an end effector **118**, one or more engagement members **122**, and one or more alignment devices **124** (e.g., a rotatable cylinder). The first, second, third, and/or fourth arms **110**, **112**, **114**, **116** and/or the end effector **118** may be coupled to, and move (e.g., rotate, pivot, translate, etc.) relative to, one another. The engagement member(s) **122** may be coupled to the end effector **118**. In some implementations, the quantity of alignment devices **124** corresponds to the quantity of nests **68** and/or tabs **70** on the plate **64** to assist with assembly of the parts (e.g., tire pressure sensors **16** and valve stems **14**) onto the wheel **12**. For example, the second robot **26** may include three alignment devices **124**.

(19) A method of assembling a wheel (e.g., wheel **12**) will now be described relative to FIGS. **1-8**. With reference to FIG. **2**, the method may begin with an operator (not shown) loading hoppers (e.g., hopper **36** and hopper **46**) with parts (e.g., tire pressure sensors **16** and valve stems **14**, respectively). One or more of the vibration mechanisms **44** may be disposed under the hoppers (e.g., hopper **36** and hopper **46**) and/or the trays (e.g., trays **38**, **40**, **48**) to convey the parts (e.g., tire pressure sensors **16** and valve stems **14**) to the container **42** (e.g., the first portion **52** and the second portion **54**). Once the parts (e.g., tire pressure sensors **16** and valve stems **14**) are received in the first portion **52** and the second portion **54**, respectively, the vibration mechanism **44** disposed under the container **42** will manipulate the parts (e.g., tire pressure sensors **16** and valve stems **14**) into a pickable position. The vision system **34** (e.g., a camera disposed above the container) may verify that the parts (e.g., tire pressure sensors **16** and valve stems **14**) are in a predetermined orientation.

(20) With reference to FIG. **3**, once the parts (e.g., tire pressure sensors **16** and valve stems **14**) are in a predetermined orientation, the first robot **24** may pick a part (e.g., valve stem **14** or tire pressure sensor **16**) from the container **42** and move the part to a lube/soap station (not shown) to apply lubrication (e.g., soap) to at least a portion of the part.

(21) As illustrated in FIG. **4**, the first robot **24** may move the part (e.g., valve stem **14** or tire pressure sensor **16**) to the staging station **20** and load the part into a nest (e.g., nest **68**) and/or attachment fixture **67** of one of the docks **58** (e.g., first dock **58-1**). In some implementations, the part is secured to the nest **68** in a press-fit configuration. The first robot **24** may repeat the foregoing steps until parts (e.g., valve stem **14** or tire pressure sensor **16**) are disposed within all (e.g., three) of the nests and/or attachment fixtures of the first dock **58-1**. While the system **10** and method are generally shown and described herein as including, or otherwise utilizing, the first robot **24** to load the parts into the nests **68**, it will be appreciated that the system and method may

include, or otherwise utilize, other means (e.g., a human) to load the parts into the nests.

(22) With reference to FIG. **5**, after the first robot **24** loads one or more (e.g., all) of the nests **68** with parts (e.g., valve stems **14** and/or tire pressure sensors **16**), the second robot **26** may return one of the plates **64**, including one or more empty nests **68**, into a dock (e.g., second dock **58-2**) and remove another of the plates **64**, including one or more nests **68** having loaded parts, from another dock (e.g., first dock **58-1**). For example, a second engagement member **122-2** of the engagement members **122** may simultaneously release one of the plates **64** onto the second dock **58-2** while a first engagement member **122-1** of the engagement members **122** removes another one of the plates **64** from the first dock **58-1**.

(23) With reference to FIGS. **6** and **7**, after the second robot **26** removes a plate (e.g., the plate **64**) from a dock (e.g., the first dock **58-1**), the second robot **26** may move the plate and the parts from the dock to the assembly station **22** and install the part onto one or more wheels (e.g., wheels **12**). In some implementations, the vision system **90**, including the camera **96**, may identify the location of a stem hole **126** on the wheel and transmit the location of the stem hole **126** to the second robot **26**. The robot **26** may position itself to install a first part (e.g., valve stem **14**) onto a first wheel **12-1** and subsequently reposition itself to install a second part (e.g., tire pressure sensor **16**) onto a second wheel **12-2**.

(24) At the assembly station, a stemming system **100** may install one or more parts (e.g., valve stems **14** or tire pressure sensors **16**) onto a wheel. For example, during the installation of a part (e.g., valve stems **14** or tire pressure sensors **16**) onto a wheel, the alignment device **124** may engage the wheel **12** and apply a force F**1** extending towards the conveyor **86**. An actuator (not shown) may apply a force F**2** on the end effector **118** causing the part to engage the wheel **12**. The force F**2** may be opposite the force F**1**, thereby ensuring that the wheel remains stationary during the installation process. In some implementations, the actuator is controlled by precision regulated air. The system **10** may include one or more sensors (e.g., a load cell, linear transducer and/or accelerometer) to determine whether the part (e.g., valve stems **14** or tire pressure sensors **16**) is properly installed in the wheel **12**. A further discussion of the stemming system **100**, including various configurations and functions thereof, may be found in commonly owned U.S. Pat. No. 11,472,240, entitled “System and Method for Stemming a Wheel,” which is hereby incorporated by reference in its entirety.

(25) With reference to FIGS. **6-8**, after installation of the part(s) (e.g., valve stems **14** or tire pressure sensors **16**) onto the wheel, the rollers **92** may move the wheel along the conveyor **86**. Hard stops and centering devices (e.g., alignment pins **88**) can maintain a location and/or alignment of the wheel relative to the conveyor **86**.

(26) After the second robot **26** removes a plate (e.g., the plate **64**) from a dock (e.g., the first dock **58-1**), and/or while the second robot **26** is installing a first part (e.g., valve stem **14**) onto the first wheel **12-1** or a second part (e.g., tire pressure sensor **16**) onto the second wheel **12-2**, the first robot **24** may simultaneously load one or more (e.g., all) of the nests **68** of another dock (e.g., the second dock **58-2**) with parts (e.g., valve stems **14** and/or tire pressure sensors **16**) as previously described.

(27) The dual-staged (e.g., two or more stations, such as staging station **20** and assembly station **22**, and/or two or more docks, such as dock **58-1** and dock **58-2**) system and method described herein allows for (a) simultaneous (i) assembly of parts (e.g., valve stems **14** or tire pressure sensors **16**) onto wheels with the second robot **26** and (ii) loading of parts (e.g., valve stems **14** or tire pressure sensors **16**) onto docks (e.g., docks **58**) with the first robot **24**, as illustrated in FIGS. **1** and **8**, and (b) simultaneous (i) placement of plates (e.g., plate **64**) onto docks (e.g., first dock **58-1**) and (ii) removal of plates (e.g., plate **64**) from docks (e.g., second dock **58-2**), thereby increasing the efficiency of the wheel assembly method.

(28) The following Clauses provide an exemplary configuration for a wheel assembly system and related methods, as described above.

(29) Clause 1: A system comprising: a docking station including a first dock and a second dock, the first dock configured to removably-receive a first plate configured to removably-receive a first plurality of parts, the second dock configured to removably-receive a second plate configured to removably-receive a second plurality of parts; and a robot including an end effector configured to engage the first plate and the second plate.
(30) Clause 2: The system of clause 1, wherein the first dock includes a first attachment fixture and a second attachment fixture, the first attachment fixture configured to receive a first part of the first plurality of parts, the second attachment fixture configured to receive a second part of the first plurality of parts.
(31) Clause 3: The system of clause 2, wherein the first dock includes a base and a pair of arms extending from the base.
(32) Clause 4: The system of clause 3, wherein the first dock includes a sensor coupled to the base and configured to sense a location of the first part.
(33) Clause 5: The system of any of clauses 1 through 4, wherein the end effector includes a first engagement member and a second engagement member, the first engagement member configured to engage the first plate, the second engagement member configured to engage the second plate when the first engagement member engages the first plate.
(34) Clause 6: The system of any of clauses 1 through 5, further comprising the first plate including a first peripheral tab and a second peripheral tab, wherein the first dock includes a first attachment fixture configured to receive the first peripheral tab, and a second attachment fixture configured to receive the second peripheral tab.
(35) Clause 7: The system of clause 6, wherein the robot further comprises a first alignment device opposing the first peripheral tab and a second alignment device opposing the second peripheral tab.
(36) Clause 8: The system of any of clauses 1 through 7, wherein the first plurality of parts includes a first type of part, and the second plurality of parts includes a second type of part different than the first type of part.
(37) Clause 9: A method comprising: positioning, with a first robot, a first part on a first plate disposed at a docking station; securing a second robot to a second plate disposed at the docking station; installing, with the second robot, a second part on a wheel; releasing the second robot from the second plate at the docking station; and securing the second robot to the first plate at the docking station.
(38) Clause 10: The method of clause 9, wherein installing, with the second robot, the second part on the wheel is simultaneous with positioning, with the first robot, the first part on the first plate.
(39) Clause 11: The method of any of clauses 9 through 10, wherein securing the first plate to the second robot is simultaneous with releasing the first plate from the second robot.
(40) Clause 12: The method of any of clauses 9 through 11, wherein installing, with the second robot, the second part on the wheel includes: applying, with the second robot, a first force in a first direction on the wheel; and applying, with the second robot, a second force in a second direction on the wheel, wherein the second direction is opposite the first direction.
(41) Clause 13: The method of any of clauses 9 through 12, further comprising sensing a location of the first part.
(42) Clause 14: The method of any of clauses 9 through 13, further comprising installing, with the second robot, a third part on the wheel.
(43) Clause 15: The method of any of clauses 9 through 14, further comprising positioning, with the first robot, a fourth part on the first plate disposed at the docking station simultaneous with installing, with the second robot, the third part on the wheel.
(44) Clause 16: The method of any of clauses 9 through 14, wherein the first part is a first type of part, and the second part is a second type of part different than the first type of part.
(45) Clause 17: A docking assembly comprising: a first dock; a first plate removably-disposed on the first dock, the first plate including a first nest and a second nest, the first nest and the second

nest each configured to removably-receive a first part and a second part; a second dock; and a second plate removably-disposed on the second dock, the second plate including a third nest and a fourth nest, the third nest and the fourth nest each configured to removably-receive the first part and the second part.

(46) Clause 18: The docking assembly of claim 17, wherein: the first plate includes a first peripheral tab and a second peripheral tab, the first nest is disposed on the first peripheral tab, and the second nest is disposed on the second peripheral tab.

(47) Clause 19: The docking assembly of any of clauses 17 through 18, further comprising a first sensor configured to sense a position of the first nest, and a second sensor configured to sense a position of the second nest.

(48) Clause 20: The docking assembly of clause 19, wherein the first dock includes a first attachment fixture having a first base and a first pair of arms extending from the first base, and wherein the first sensor is disposed on the first base, and wherein the second dock includes a second attachment fixture having a second base and a second pair of arms extending from the second base, and wherein the second sensor is disposed on the second base.

(49) The terminology used herein is for the purpose of describing particular exemplary configurations only and is not intended to be limiting. As used herein, the singular articles "a," "an," and "the" may be intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms "comprises," "comprising," "including," and "having," are inclusive and therefore specify the presence of features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. Additional or alternative steps may be employed.

(50) When an element or layer is referred to as being "on," "engaged to," "connected to," "attached to," or "coupled to" another element or layer, it may be directly on, engaged, connected, attached, or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being "directly on," "directly engaged to," "directly connected to," "directly attached to," or "directly coupled to" another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., "between" versus "directly between," "adjacent" versus "directly adjacent," etc.). As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items.

(51) The terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections. These elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as "first," "second," and other numerical terms do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the example configurations.

(52) The foregoing description has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular configuration are generally not limited to that particular configuration, but, where applicable, are interchangeable and can be used in a selected configuration, even if not specifically shown or described. The same may also be varied in many ways. Such variations are not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

Claims

1. A system comprising: a docking station including: a first dock; a first plate configured to be removably-received by the first dock and configured to removably-receive a first plurality of parts, wherein: the first plate includes a first peripheral tab and a second peripheral tab, and the first dock includes (i) a first attachment fixture configured to receive the first peripheral tab and (ii) a second attachment fixture configured to receive the second peripheral tab; and a second dock configured to removably-receive a second plate configured to removably-receive a second plurality of parts; and a robot including an end effector configured to engage the first plate and the second plate.
2. The system of claim 1, wherein: the first attachment fixture is configured to receive a first part of the first plurality of parts, and the second attachment fixture is configured to receive a second part of the first plurality of parts.
3. The system of claim 2, wherein the first attachment fixture includes a base and a pair of arms extending from the base.
4. The system of claim 3, wherein the first dock-first attachment fixture includes a sensor coupled to the base and configured to sense a location of the first part.
5. The system of claim 1, wherein; the end effector includes a first engagement member and a second engagement member, the first engagement member is configured to engage the first plate, and the second engagement member is configured to engage the second plate when the first engagement member engages the first plate.
6. The system of claim 1, wherein the robot further includes a first alignment device opposing the first peripheral tab and a second alignment device opposing the second peripheral tab.
7. The system of claim 1, wherein the first plurality of parts includes a first type of part, and the second plurality of parts includes a second type of part different than the first type of part.